Lab 3: SIS/SIRS Epidemic Simulation

Goal:
Learn how epidemic models like SIS and SIRS evolve on a network using `graph-tool`.

Steps:

1. **Read:**
   Go through this documentation:
   SIRSState — graph-tool docs

2. **Understand and Replicate this code on Google Colab:**

```
from graph_tool.all import *
import matplotlib.pyplot as plt

g = gt.collection.data["pgp-strong-2009"]
state = gt.SISState(g, beta=0.01, gamma=0.007)
X = []

for t in range(1000):
    state.iterate_sync()
    X.append(state.get_state().fa.sum())

plt.figure(figsize=(6,4))
plt.plot(X)
plt.xlabel("Time")
plt.ylabel("Infectious nodes")
plt.tight_layout()
plt.show()
```

3. **Try it yourself:**
   a) Change the values of `beta` and `gamma` to see how the curve changes.
   b) (Optional) Try using `gt.SIRSState` instead of `gt.SISState` and plot S, I, R.
4. **Submit:**
   a) Your Colab notebook link
   b) The resulting plot and a short note on what you observed.